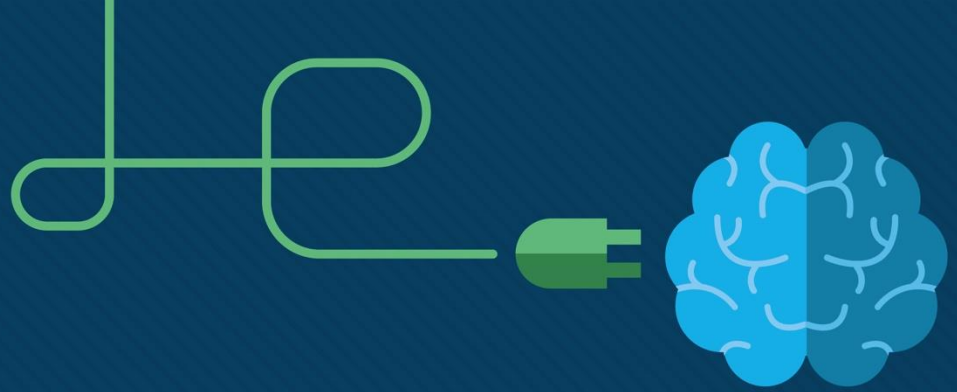

Module 6
Network Management

ITNET03

Advanced Routing and Switching

Module Objectives

Module Title: Network Management

Module Objectives:

- Use network discovery tools to map a network topology
- Implement correct timestamping and logging of network events through event logging .
- Perform remote network management through SNMP
- Perform basic network device file management
- Perform password recovery on a router

Module References:

- CCNAv7 ENSA – Module 10

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6.1 Introduction to Network Management

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Introduction

- Once a network is deployed, it is essential that it is correctly managed to ensure network availability. Basic tasks include:
  - Monitoring the network for usage patterns, data flows and issues
  - Managing device OS and configurations
- Why?
  - Keep informed of failures or irregularities in the network through notifications
    - Report and diagnose issues
    - Ensures uptime
    - Remediate disasters by detecting impending failures before they happen
  - Helps plan for change
    - Track usage trends and network capacity
    - Information collection and review allows growth projection
  - Have the ability to restore device operations in case of failure
    - Backing up and restore critical device files when needed

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Network Management Tools

- As a network grows, monitoring tools are essential to simplify administrative tasks and make network management more feasible and practical
- Monitoring and management protocols
  - Cisco Discovery Protocol (CDP) and Link Layer Discovery Protocol (LLDP)
  - Network Time Protocol (NTP)
  - Syslog
  - Simple network management protocol (SNMP)
- Managing devices also involves backing up important files and having the capability to performing password recovery when needed

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6.2 Device Discovery Tools

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CDP and LLDP

- Device discovery tools are used to gather information about devices which share the same data link and are often used for network documentation and automatic neighbor discovery.
  - **Cisco Discovery Protocol (CDP)** is a Cisco proprietary used exclusively by Cisco devices
  - **Link Layer Discovery Protocol (LLDP)** is a vendor-neutral neighbor discovery protocol and works between devices from different manufacturers
- When enabled, devices send periodic advertisements about their identity and capabilities to neighbors on the same physical link.

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Configure CDP or LLDP

- For Cisco devices, CDP is enabled by default, but LLDP is also supported. Both protocols may be enabled / disabled globally on a device or on a per interface basis.

	CDP	LLDP
Enable globally	`Router(config)#cdp run`	`Router(config)#lldp run`
Enable per interface	`Router(config-if)#cdp enable`	`Router(config-if)#lldp transmit` `Router(config-if)#lldp receive`
Disable globally	`Router(config)#no cdp run`	`Router(config)#no lldp run`
Disable per interface	`Router(config-if)#no cdp enable`	`Router(config-if)#no lldp transmit` `Router(config-if)#no lldp receive`
Verify global status	`Router>show cdp`	`Router>show lldp`
Verify interface status	`Router>show cdp interface`	-

- Note: It is recommended that CDP/LLDP be disabled when not in use to avoid leaking sensitive device information

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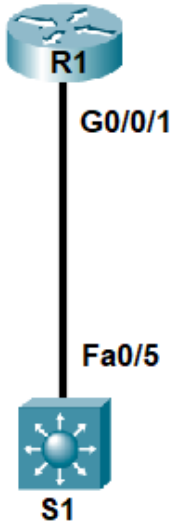
Discover Network Topology

- With enabled on the network, the **show cdp neighbors** or **show lldp neighbors** command can be used to determine the network layout




```
R1# show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID          Local Intrfce  Holdtme   Capability     Platform    Port ID
S1                 Gig 0/0/1      179        S I           WS-C3560-   Fas 0/5
```

```
S1# show lldp neighbors
Capability codes:
    (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
    (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other
Device ID      Local Intf     Hold-time      Capability     Port ID
R1             Fa0/5          117            R              Gi0/0/1
Total entries displayed: 1
```

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Discover Device Details

- The **show cdp neighbors detail** or **show lldp neighbors detail** command can be used gather detailed information about device hardware, firmware and IP settings

```
R1# show cdp neighbors detail
-------------------------
Device ID: S1
Entry address(es):
  IP address: 192.168.1.2
Platform: cisco WS-C3560-24TS, Capabilities:
Switch IGMP
Interface: GigabitEthernet0/0/1, Port ID (outgoing
port): FastEthernet0/5
Holdtime : 136 sec

Version :
Cisco IOS Software, C3560 Software (C3560-
ADVIPSERVICESK9-M), Version 12.2(37)SE1, RELEASE
SOFTWARE (fc1)
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Thu 05-Jul-07 22:22 by pt_team

advertisement version: 2
Duplex: full
```

```
S1# show lldp neighbors detail
------------------------------------------------
Chassis id: 848a.8d44.49b0
Port id: Gi0/0/1
Port Description: GigabitEthernet0/0/1
System Name: R1
System Description:
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-
M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2019 by Cisco Systems, Inc.
Compiled Thu 22-Aug-19 18:09 by mcpre

Time remaining: 111 seconds
System Capabilities: B,R
Enabled Capabilities: R
Management Addresses - not advertised
Physical media capabilities:
    100baseT(FD)
    1000baseT(HD)
Media Attachment Unit type: 10
Vlan ID: 1
```

6.3 Event Logging

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Time and Logging Services

- In a production network, a network administrator will not always be on hand when network events such as device status changes or security violations occur
- Logging services allow devices to record events either locally or to a remote central repository when they occur to be later reviewed by the network administrator
- Time services are used simultaneously with logging to allow network events to be appropriately timestamped when they are recorded
- Network devices will often have a software clock that can be manually set:

```
R1# clock set 20:36:00 nov 15 2019
R1#
*Nov 15 20:36:00.000: %SYS-6-CLOCKUPDATE: System clock has been updated from
21:32:31 UTC Fri Nov 15 2019 to 20:36:00 UTC Fri Nov 15 2019, configured from
console by console.
```

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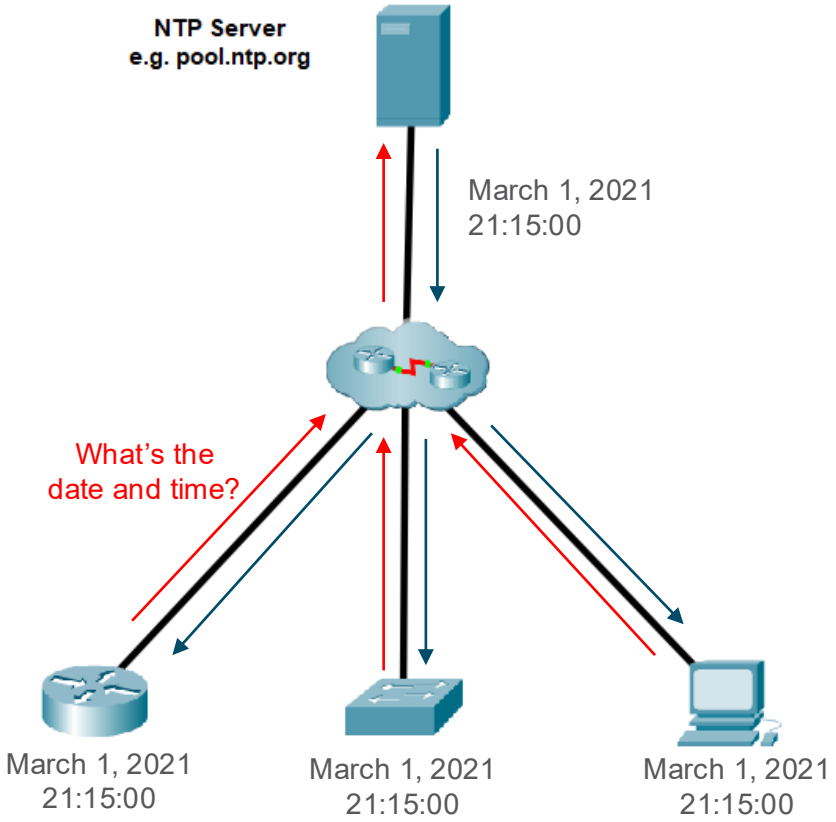
Time and Calendar Services

- It is important to synchronize the time across all devices on the network. When the time is not synchronized between devices, it will be impossible to determine the order of the events and the cause of an event.
- As a network grows, it becomes difficult to ensure that all infrastructure devices are operating with synchronized time using the manual method.
- The **Network Time Protocol (NTP)** allows network devices to synchronize their time settings with a server to ensure more consistent time settings across different devices.



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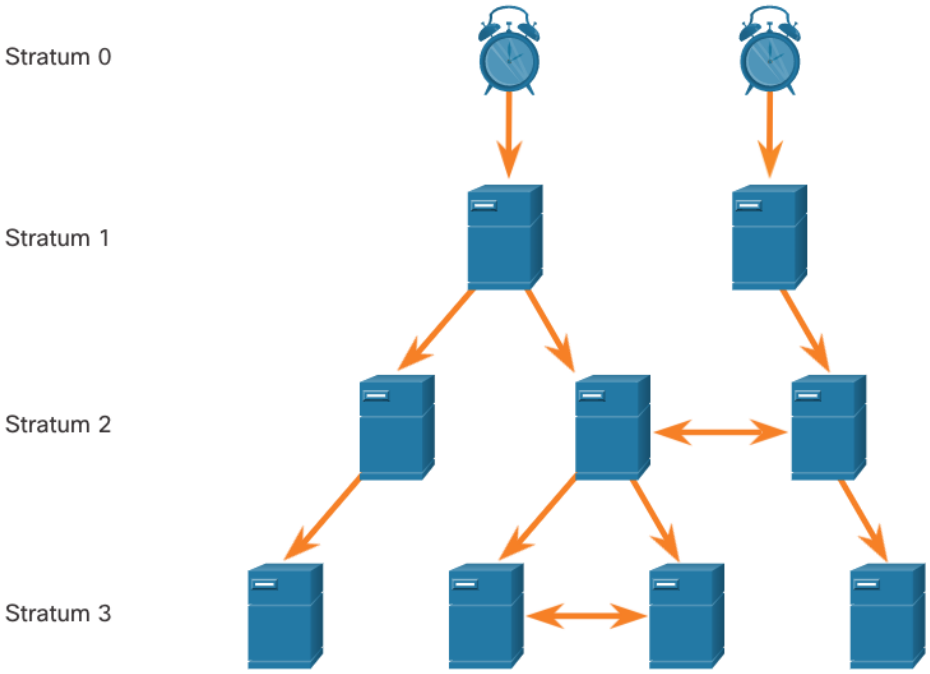
NTP Operation

- NTP networks use a hierarchical system of time sources.
- Each level in this hierarchical system is called a **stratum** and the synchronized time is distributed across the network using NTP packets
- The stratum level is defined as the number of hop counts from the authoritative source..
- The max hop count is 15. Stratum 16, the lowest stratum level, indicates that a device is unsynchronized.



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NTP Operation (Cont.)

- **Stratum 0:** These authoritative time sources are high-precision timekeeping devices assumed to be accurate and with little or no delay associated with them.
- **Stratum 1:** Devices that are directly connected to the authoritative time sources. They act as the primary network time standard.
- **Stratum 2 and Lower:** Stratum 2 servers are connected to stratum 1 devices through network connections. Stratum 2 devices, such as NTP clients, synchronize their time by using the NTP packets from stratum 1 servers. They could also act as servers for lower stratum.

Time servers on the same stratum level can be configured to act as a peer with other time servers on the same stratum level for backup or verification of time.

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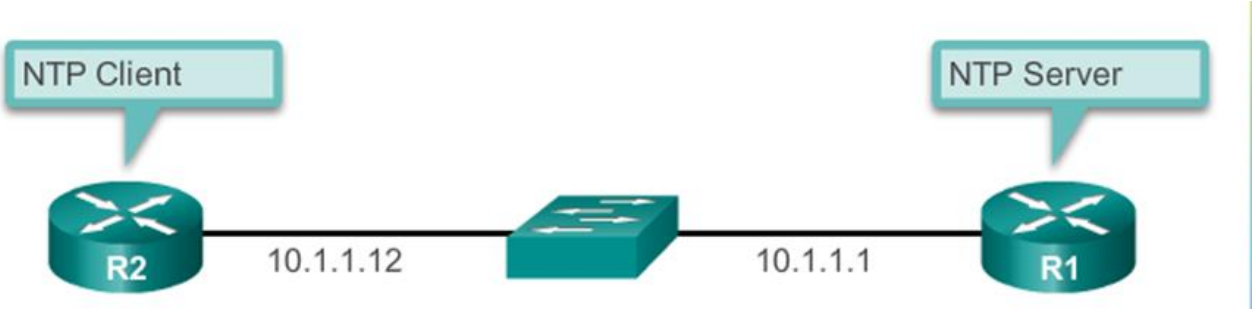
Configure NTP

- An NTP server can be a publicly available server on the Internet or a private master clock
- Devices that will synchronize with an NTP server need to be configured as an NTP client:

```
Router(config)#ntp server srver_ip
```

- A local router or switch can be used as an NTP server for the rest of the network using its own system clock as reference

```
Router(config)#ntp master stratum
```




```
R2(config)#ntp server 10.1.1.1
```

```
R1(config)#ntp master 1
```

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Verify NTP

show clock detail – verify current time and time source

```
R1# show clock detail
21:01:34.563 UTC Fri Nov 15 2019
Time source is NTP
```

show ntp associations - list the NTP time sources of a device and their respective stratum levels

```
R2# show ntp associations
address         ref clock       st when poll reach delay offset disp
*~10.1.1.1      127.127.1.1     1  12   64   377   1.066 13.616 3.840
• sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

show ntp status – show the NTP synchronization status of a device and its own stratum level

```
R1# show ntp status
Clock is synchronized, stratum 2, reference is 10.1.1.1
nominal freq is 119.2092 Hz, actual freq is 119.2088 Hz, precision is 2**17
(output omitted
```

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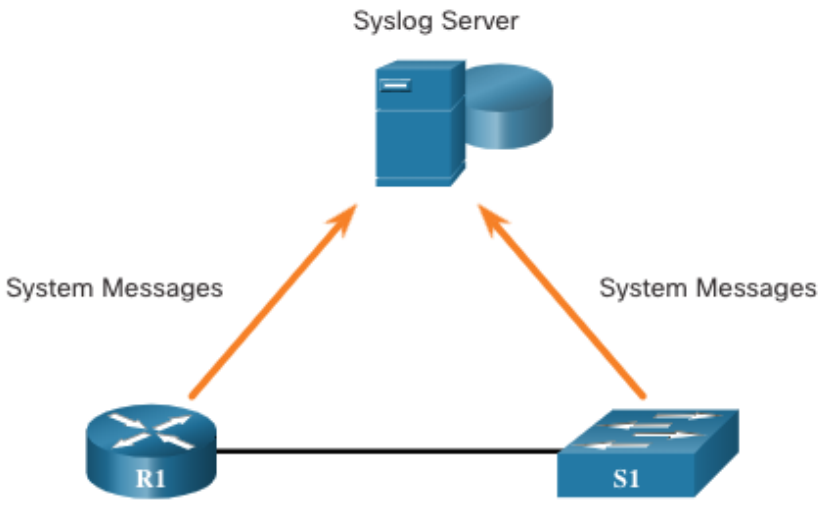
Introduction to Syslog

- Syslog is a protocol used for network event logging
- Can send event notification messages across IP networks to event message collectors using UDP port 514
- The syslog logging service provides three primary functions, as follows:
  - The ability to gather logging information for monitoring and troubleshooting
  - The ability to select the type of logging information that is captured
  - The ability to specify the destinations of captured syslog messages



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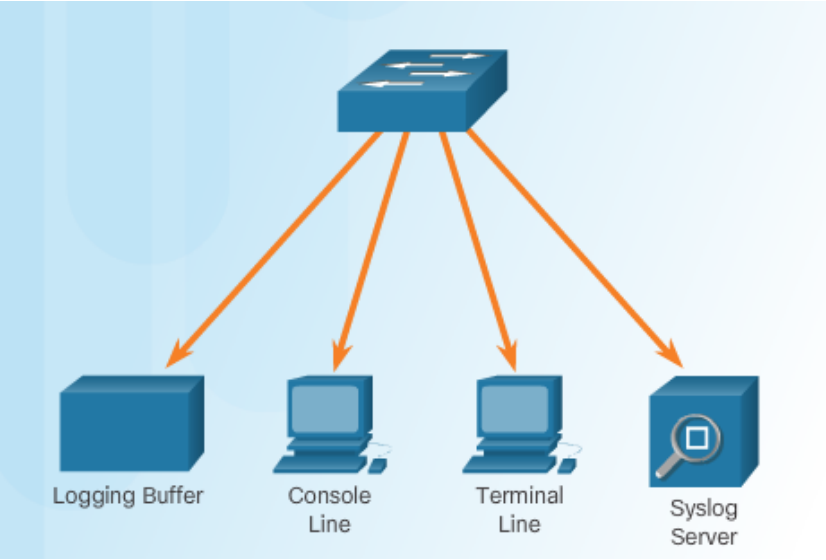
Syslog Operation

- Syslog protocol starts by sending system messages and debug output to a local logging process internal to the device.
- How the logging process manages these messages and outputs is based on device configurations. Destinations for syslog messages include:
  - Terminal / console line – outputted to the CLI and can be viewed only in realtime
  - Syslog server- sent across the network to an external server which can store messages and generate various reports.
  - Internal buffer – stored in device RAM that can later be reviewed through the CLI of the device.



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Syslog Message Format

- Cisco devices produce syslog messages as a result of network events
- Cisco syslog messages use the following default format:

seq no:timestamp: %facility-severity-MNEMONIC:description

- Example:

```
*Feb 23, 11:17:45.1717: %LINK-5-CHANGED: Interface Vlan1, changed state to administratively down
```

Field	Description
Seq no	Stamps log messages with a sequence number only if the service sequence-numbers option is enabled
timestamp	Date and time of message or event. Appears if the service timestamps option is enabled
facility	Device component / protocol / module that generated the message
severity	Numerical code representing criticality of message
MNEMONIC	Text string that uniquely describes the message – often used to facilitate message filtering
description	Text string containing detailed information about the reported event

Syslog Facilities

```
Feb 23, 11:17:45.1717: %LINK-5-CHANGED: Interface Vlan1, changed state to administratively down
```

- Syslog facilities are service identifiers that identify which device hardware component, software component, protocol or module generated the event
- Often used to categorize system state data for error and event message reporting.
- Logging facility options that are available are specific to the networking device. Some common syslog message facilities reported on Cisco IOS routers include:
  - IP
  - OSPF protocol
  - SYS operating system
  - IP security (IPsec)
  - Interface IP (IF)

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Syslog Message Severity

```
Feb 23, 11:17:45.1717: %LINK-5-CHANGED: Interface Vlan1, changed state to administratively down
```

Severity Name	Severity Level	Explanation
Emergency	Level 0	System Unusable
Alert	Level 1	Immediate Action Needed
Critical	Level 2	Critical Condition
Error	Level 3	Error Condition
Warning	Level 4	Warning Condition
Notification	Level 5	Normal, but Significant Condition
Informational	Level 6	Informational Message
Debugging	Level 7	Debugging Message

- Severity levels range form 0-7, with lower values representing more urgent messages
- Each syslog level has its own meaning:
  - **Warning Level 4 - Emergency Level 0**: Error messages about software or hardware malfunctions; functionality of the device is affected.
  - **Notification Level 5**: The notifications level is for normal events. e.g. interface state transitions, system restart
  - **Informational Level 6**: A normal information message that does not affect device functionality.
  - **Debugging Level 7**: This level indicates that the messages are output generated from issuing various **debug** commands.

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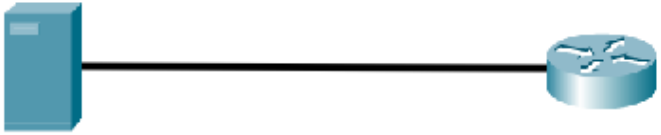
Configure Syslog

To configure syslog operations on a device

Task	Command
Enable message timestamping	`Router(config)# service timestamps log datetime msec`
Send log messages to a server	`Router(config)# logging serv_ip`
Set logging trap level *includes configured level + more critical messages – e.g.logging level 5 will include severity levels 0-5	`Router(config)# logging trap level`



Syslog Server

192.168.1.100

```
R1(config)#service timestamps log datetime msec
R1(config)#logging 192.168.1.100
R1(config)#logging trap 5
```

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Verify Syslog

Use the show logging command to verify syslog operation on a device and view its logged events if internal buffer logging is enabled

```
R1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
          0 flushes, 0 overruns, xml disabled, filtering disabled)

--output omitted--

    Console logging: level debugging, 17 messages logged, xml disabled,
          filtering disabled
    Monitor logging: level debugging, 17 messages logged, xml disabled,
          filtering disabled
    Buffer logging:  level debugging, 17 messages logged, xml disabled,
          filtering disabled

    Logging Exception size (4096 bytes)
    Count and timestamp logging messages: disabled
    Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
    Trap logging: level informational, 17 message lines logged
        Logging to 192.168.1.100  (udp port 514,  audit disabled,
              authentication disabled, encryption disabled, link up),
              2 message lines logged,
              0 message lines rate-limited,
              0 message lines dropped-by-MD,
              xml disabled, sequence number disabled
              filtering disabled
Log Buffer (32768 bytes):
*Feb 23, 11:52:28.5252: %LINK-5-CHANGED: Interface Vlan1, changed state to up
*Feb 23, 11:52:28.5252: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1,
```

6.4 Centralized Device Management

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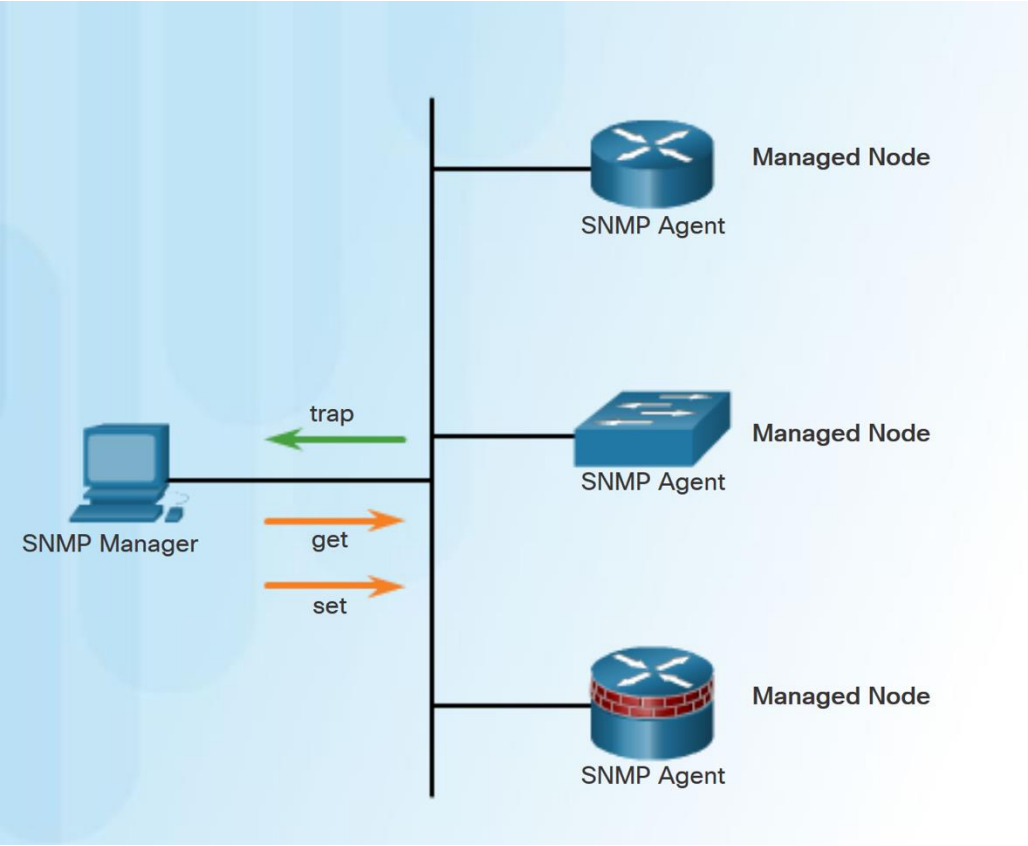
Introduction to SNMP

- Simple Network Management Protocol (SNMP) enables network administrators to monitor and manage network nodes.
- A standard protocol for
  - collecting and organizing information about managed devices in a network
  - Modifying device information (i.e. configurations) to control device operations
- Defines how information is transferred between manager and managed devices



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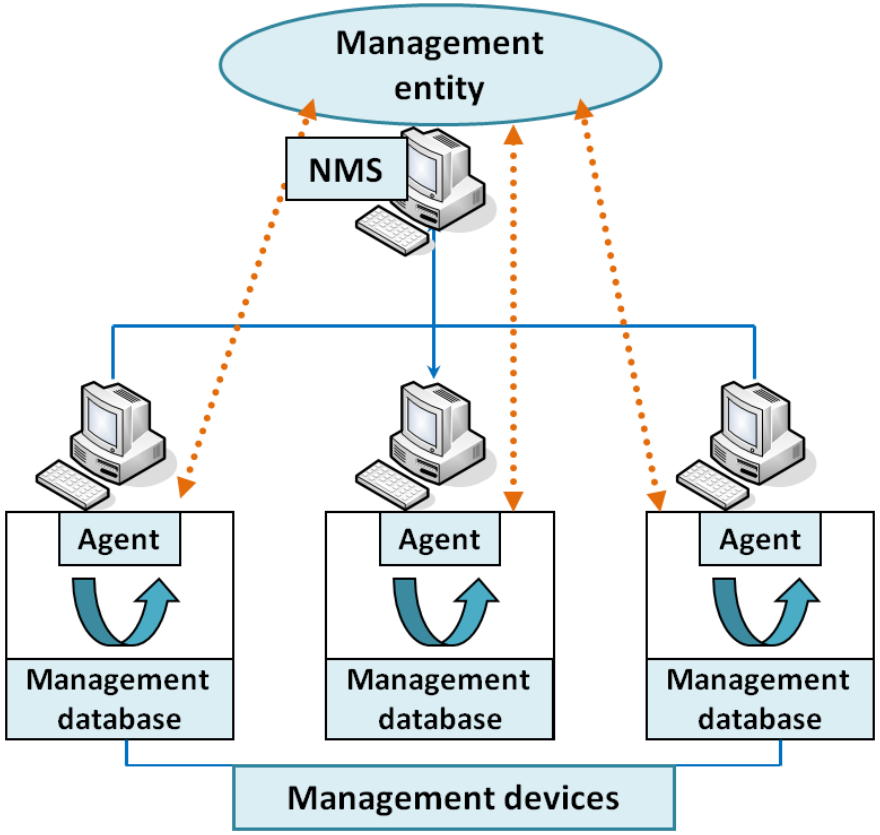
Introduction to SNMP

- The SNMP system consists of the following elements:
  - **SNMP manager-** collects information from an SNMP agent using the “get” action. Changes configurations on an agent using the “set” action.
  - **Managed Device** – Any network node that implements an SNMP interface to allow read or read/write access to node specific information
  - **Agent** - Software running on a managed device that responds to requests from the manager
  - **Management Information Base (MIB)-** a structure that stores data and operational statistics about the managed device.



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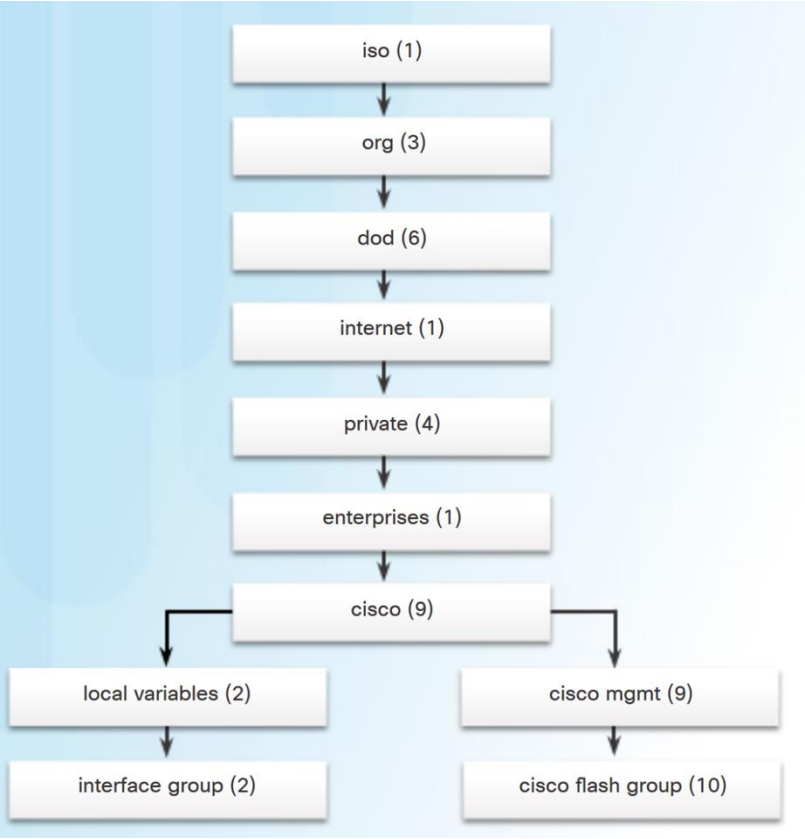
Management Information Base

- A hierarchical, pre-defined structure that stores device information as variables that can be queried or set by authenticated managers
- The MIB defines each variable as an object ID (OID).
  - OIDs uniquely identify managed objects.
  - Most devices implement RFC defined common public variables that are common across different equipment manufacturers.
  - Vendors such as Cisco can define private OIDs for vendor-specific variables.



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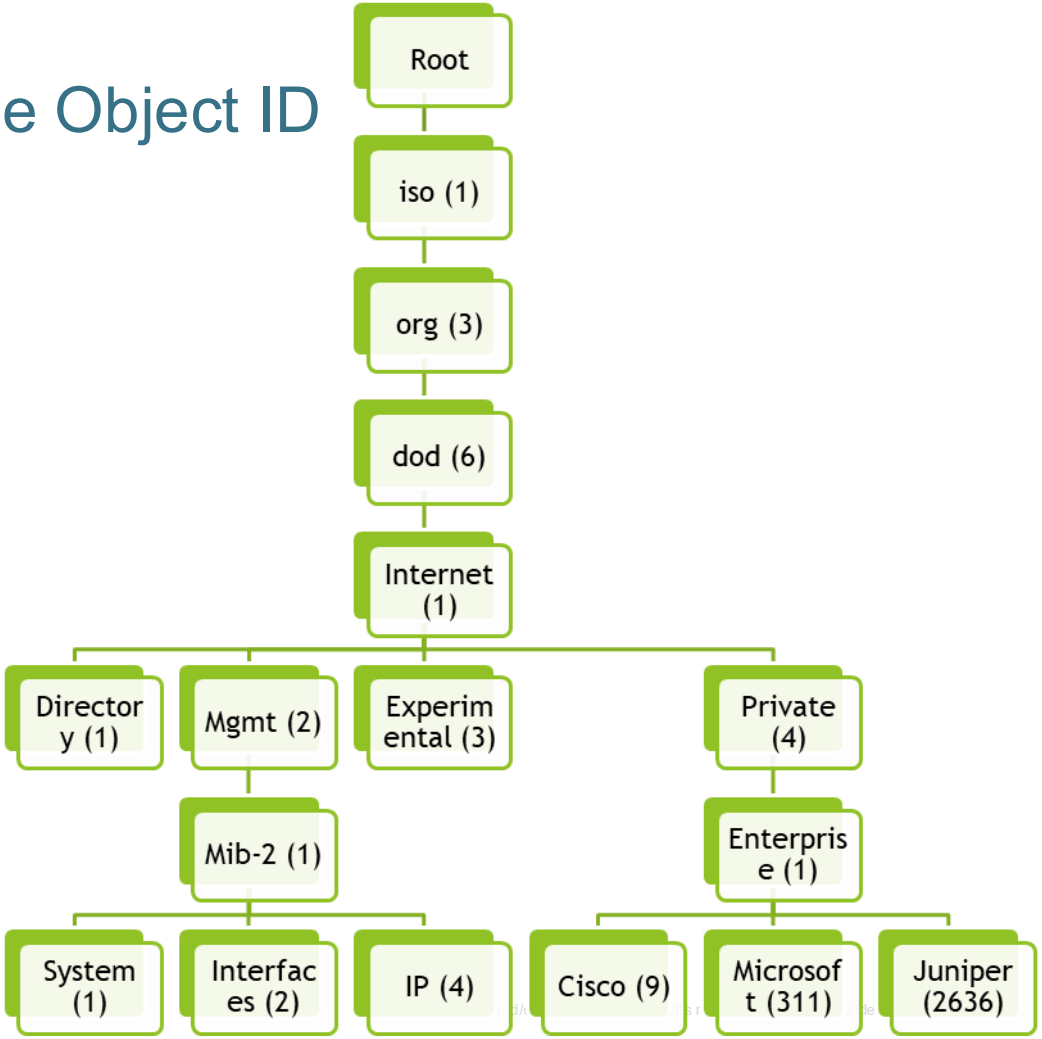
Management Information Base Object ID

- The MIB is similar in structure to a top-down hierarchical tree
  - Branches are identified by a number or a string
  - Nodes are identified by paths consisting of the top-down sequence of branch identifiers (object identifier or OID)
  - Sample OID to access the list of device interfaces
    - 1.3.6.1.2.1.2 or
    - iso.org.dod.internet.mgmt.mib-2.interfaces

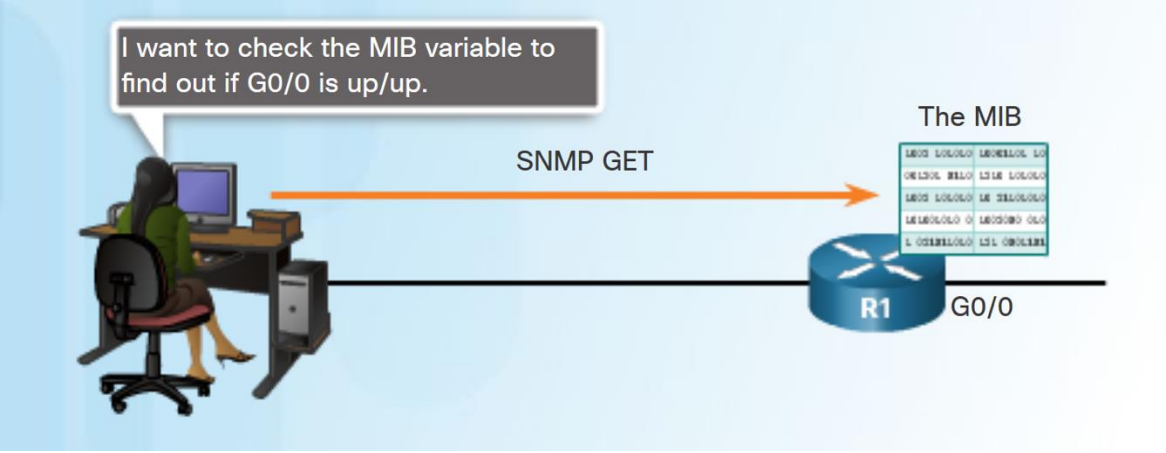


SNMP Operation

- SNMP agents that reside on managed devices collect and store information locally in each device's MIB.
  - A Network Management System (NMS) queries the SNMP agents using the GET or SET messages.
  - Agents respond to the requests using an SNMP RESPONSE message
  - The SNMP manager polls the agents and queries the MIB for SNMP agents on UDP port 161.

Messages sent by the NMS

- GET - Retrieves the value of an object
- GET-NEXT - Retrieves a value from a variable in a table sequentially
- GET-BULK - Retrieves large blocks of data (e.g. an entire table)
- SET - Modifies the value of a specific variable



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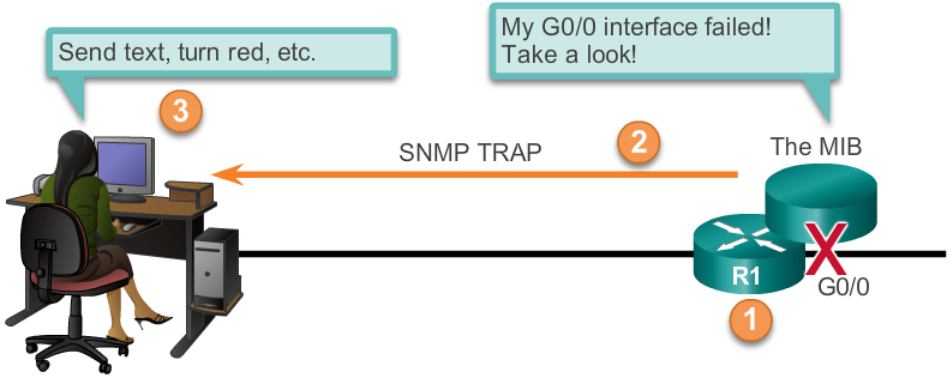
SNMP Agent Traps

- A Network Management System (NMS) can periodically poll the SNMP agents to monitor certain variables using the get request - e.g. CPU Load, Interface Traffic Load
- Polling for everything is not always ideal
  - Delay between event occurrence and next poll
  - Higher polling frequency = higher bandwidth usage
- Solution: SNMP Traps - Agents send unsolicited messages alerting the SNMP manager to a condition or event such as improper user authentication or link status.
- SNMP agents send any SNMP traps to the SNMP manager on UDP port 162.



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SNMP Versions

SNMP currently has 3 versions that can be used in a network

- Version 1
  - Initial deployment of the standard
  - Access control to MIB is implemented through plaintext community strings
  - Generally not used today
- Version 2c
  - Includes improvements in protocol performance and the Get-Bulk and Inform messages
  - Still uses plaintext community strings
- Version 3
  - Newest version that has offers authentication, encryption and access-control for security
  - Can define user groups and access policies for these groups

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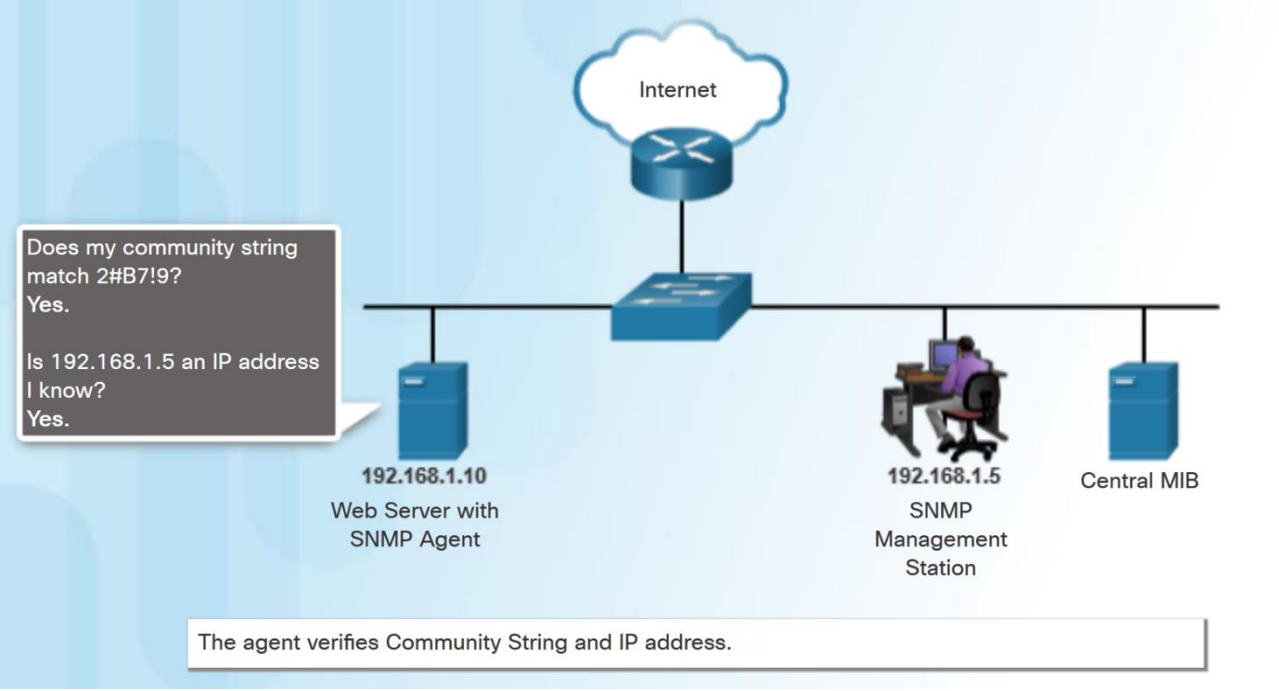
Community Strings

- SNMPv1 and SNMPv2c use community strings that control access to the MIB.
- Two types of community strings:
  - **Read-only (ro)** - Provides access to the MIB variables, but no changes can be made.
  - **Read-write (rw)** - Provides read and write access to all objects in the MIB.



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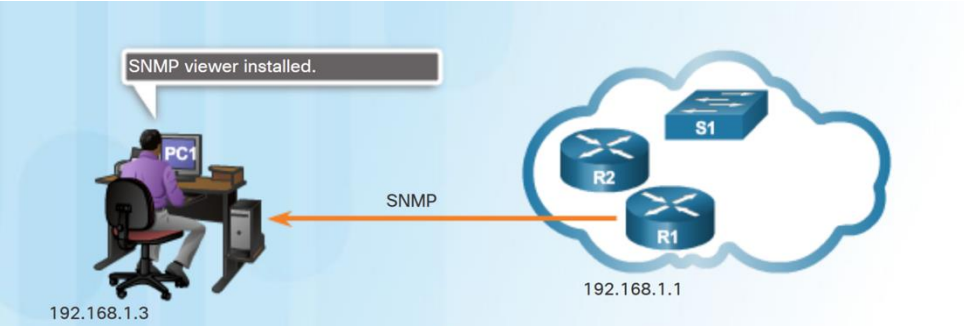
Steps for Configuring SNMP




```
R1(config)#snmp-server community COMString ro
R1(config)#snmp-server enable traps
R1(config)#snmp-server host 192.168.1.3 version 2c
     COMString
```

- Basic steps to configuring SNMP:
  1. Configure the community string and access level

     `Router(config)#snmp-server community string {ro | rw}`
  2. (Optional) Enable SNMP traps from the agent

     `Router(config)# snmp-server enable traps notif-types`
  3. (Optional) Specify the recipient of the SNMP trap operations

     `Router(config)# snmp-server host host-id [version {1 | 2c |3 }] community-string`

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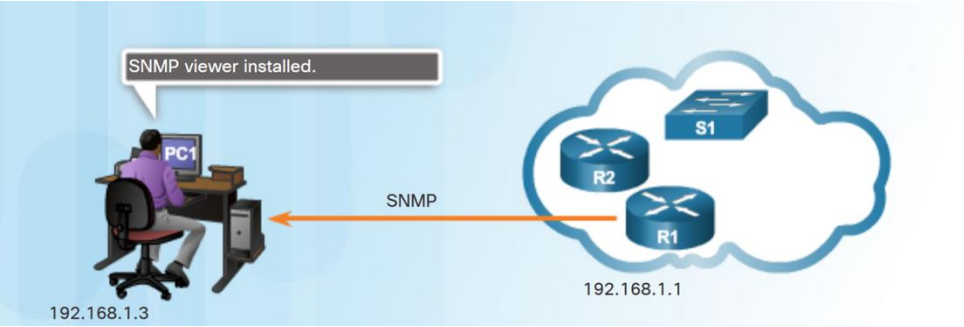
Steps for Configuring SNMP




```
R1(config)#snmp-server community COMString ro SNMP_ACL
R1(config)#snmp-server enable traps
R1(config)#snmp-server host 192.168.1.3 version 2c
     COMString
R1(config)#snmp-server location Batcave_rackA
R1(config)#snmp_server contact Bruce Wayne
R1(config)#ip access-list standard SNMP_ACL
R1(config-std-nacl)#permit host 192.168.1.3
```

- Basic steps to configuring SNMP:

4. (Optional) Document the location of the device

   `Router(config)# snmp-server location text`

5. (Optional) Document the system contact person

   `Router(config)# snmp-server contact text`

6. (Optional) Use an ACL to restrict SNMP access to NMS hosts (SNMP managers).

   `Router(config)# snmp-server community string {ro | rw} acl_id`

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Verifying SNMP Configuration

- To verify the SNMP configuration use the **show snmp** command.

```
R1#show snmp
Chassis: FTX0945W0MY
Contact: Bruce Wayne
Location: Batcave_rackA
0 SNMP packets input
    0 Bad SNMP version errors
    0 Unknown community name
    0 Illegal operation for community name supplied
    0 Encoding errors
    0 Number of requested variables
    0 Number of altered variables
    0 Get-request PDUs
    0 Get-next PDUs
    0 Set-request PDUs
    0 Input queue packet drops (Maximum queue size 1000)
7 SNMP packets output
    0 Too big errors (Maximum packet size 1500)
    0 No such name errors
    0 Bad values errors
    0 General errors
    0 Response PDUs
    7 Trap PDUs

SNMP logging: enabled
    Logging to 192.168.10.3.162, 0/10, 3 sent, 0 dropped.
```

Verifying SNMP Configuration

- Use the **show snmp community** command to show SNMP community string and ACL information.

```
R1#show snmp community

Community name: ILMI
Community Index: cisco0
Community SecurityName: ILMI
storage-type: read-only  active


Community name: COMString
Community Index: cisco1
Community SecurityName: COMString
storage-type: nonvolatile        active access-list: SNMP_ACL
```

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6.5 Router and Switch File Maintenance

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Router File Systems

- The Cisco IOS File System (IFS) allows the administrator to navigate to different directories and list the files in a directory. The administrator can also create subdirectories in flash memory or on a disk.
- The **show file systems** command, lists all available file systems on a Cisco device.
- An entry with asterisk indicates the current default file system. The pound sign (#) indicates a bootable disk. Both of these are assigned to the flash file system by default

```
Router# show file systems
File Systems:
       Size(b)       Free(b)      Type  Flags  Prefixes
             -             -    opaque     rw   system:
             -             -    opaque     rw   tmpsys:
*   7194652672    6294822912      disk     rw   bootflash: flash:
     256589824     256573440      disk     rw   usb0:
    1804468224    1723789312      disk     ro   webui:
             -             -    opaque     rw   null:
             -             -    opaque     ro   tar:
             -             -   network     rw   tftp:
             -             -    opaque     wo   syslog:
      33554432      33539983     nvram     rw   nvram:
             -             -   network     rw   rcp:
             -             -   network     rw   ftp:
             -             -   network     rw   http:
             -             -   network     rw   scp:
             -             -   network     rw   sftp:
             -             -   network     rw   https:
             -             -    opaque     ro   cns:
Router#
```

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Basic Device File System Management

- **show flash** or **dir** command can be used to list the contents of flash of a Cisco router and switch

```
Router#show flash

System flash directory:
File  Length   Name/status
  3   33591768 c1900-universalk9-mz.SPA.151-4.M4.bin
  2   28282    sigdef-category.xml
  1   227537   sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)


Router#dir
Directory of flash0:/

    3  -rw-    33591768          <no date>  c1900-universalk9-mz.SPA.151-4.M4.bin
    2  -rw-       28282          <no date>  sigdef-category.xml
    1  -rw-      227537          <no date>  sigdef-default.xml

255744000 bytes total (221896413 bytes free)
```

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Basic Device File System Management

- **`delete`** *`filesystem:filename`* command is used to erase a file

  Example:

  ```
  Switch# delete flash:vlan.dat
  ```

  or

  ```
  Switch# delete vlan.dat
  ```

- **`copy srcfilesystem:filename destfilesystem:filename`** command is used to create a copy of the source file in the specified destination

  Example:

  ```
  Router# copy running-config startup-config
  ```

  or

  ```
  Router# copy system:running-config nvram:startup-config
  ```

- Specifying only the filesystem in the copy command brings up a prompt to input the filenames to be copied. ← Only method supported in Packet Tracer

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Backing Up Device Files

- Creating copies of device OS images and configuration files is an essential task in network management to ensure that a backup is available in case of there is a need to for restoration after file corruption, accidental file deletion or configuration rollback
- Common options for backing up and restoring files:
  - Copying to/from a different filename in flash memory
  - Copying to/from an external USB drive
  - Copying to/from a remote Trivial File Transfer Protocol (TFTP) server
- Where are the files kept by default?
  - IOS image – flash
  - Startup configuration – nvram (routers and L3 switches), flash (switches)

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Using Internal Flash for Backup

- The internal flash memory of a device can be used to save a duplicate copy of the OS or startup configuration
- Example:
  - To backup the router startup configuration to flash:

```
Router#copy startup-config flash:
Destination filename [startup-config]? config-backup-2021-02-24

613 bytes copied in 1.796 secs  secs (1473 bytes/sec)
```

  - To restore the router startup configuration from a backup copy in flash:

```
Router#copy flash: startup-config
Source filename []? config-backup-2021-02-24
Destination filename [startup-config]?
[OK]

613 bytes copied in 1.796 secs  secs (1473 bytes/sec)
```

- Note: If the file will be backed up into its default filesystem, use a different filename to save the backup to avoid overwriting the original file

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Using a USB Drive for Backup

- The Universal Serial Bus (USB) storage feature enables certain models of Cisco routers to support USB flash drives to store files externally.

- Example:
  - To backup the router startup configuration to a USB drive:

```
Router#copy startup-config usbflash0:
Destination filename [startup-config]? config-backup-2021-02-24

613 bytes copied in 0.416 secs (1473 bytes/sec)
```

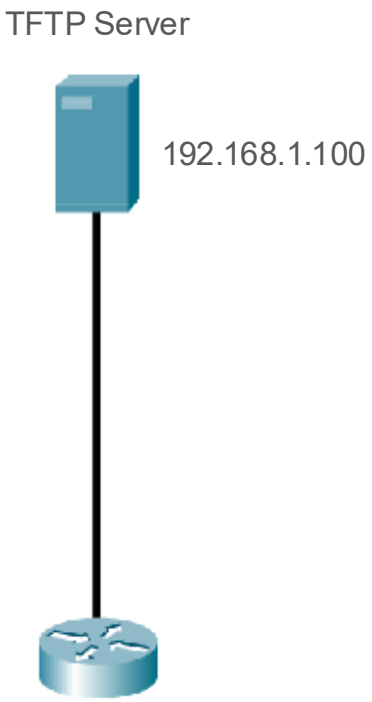
  - To restore the router startup configuration from a backup copy in a USB drive:

```
Router#copy usbflash0: startup-config
Source filename []? config-backup-2021-02-24
Destination filename [startup-config]?
[OK]

613 bytes copied in 0.416 secs (1473 bytes/sec)
```

Using a Remote Server

- **Trivial File Transfer Protocol (TFTP)** is a lightweight file UDP-based file transfer protocol often used to allow networked devices to boot from a remotely stored OS image
- As a network grows, IOS Software images and configuration files can be stored on a central TFTP server. This helps to control the number of IOS images and the revisions to those IOS images, as well as the configuration files that must be maintained.
- To use TFTP or file backup, a router or switch must have an IP address and must be able to reach the server over the network



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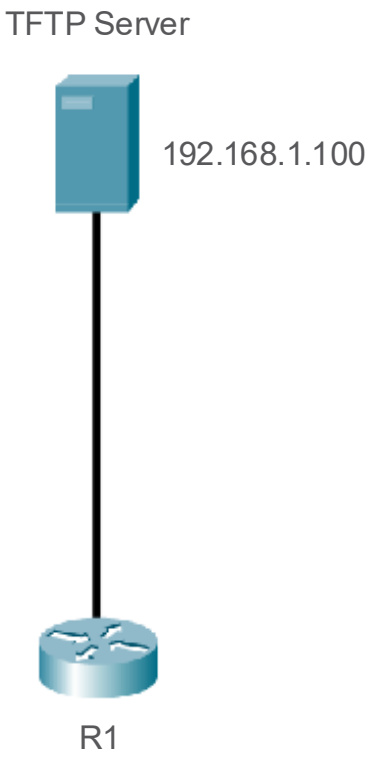
Using a Remote Server

- To backup a file to a TFTP server:

```
R1#copy startup-config tftp:
Address or name of remote host []? 192.168.1.100
Destination filename [startup-config]? R1_config-2021-02-24
Write file R1_config-2021-02-24 to 192.168.100.100? [confirm]

Writing R1-Jan-2019 !!!!!! [OK]

622 bytes copied in 0 secs
```

- To restore a file from a TFTP server

```
R1#copy tftp: startup-config
Address or name of remote host []? 192.168.1.100
Source filename []? R1_config-2021-02-24
Destination filename [startup-config]?

Accessing tftp://192.168.1.100/R1_config-2021-02-24...
Loading R1_config-2021-02-24 from 192.168.1.100: !!!!!! [OK]

622 bytes copied in 0 secs
```



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IOS Image Management

- By default, a Cisco router or switch router defaults to loading the first valid Cisco IOS image in flash memory and runs it.
- A router or switch can store multiple copies / versions of its OS in flash memory
  - The **`boot system`** *`filesystem:image_name`* command is used to specify which OS image to load on startup
  - Several **boot system** commands can be entered in sequence to specify boot image priority and provide a fault-tolerant boot plan.
- Example

OS images

```
R1# dir
Directory of flash:/

    3  -rw-    33591768          <no date>  c1841-advipservicesk9-mz.124-15.T1.bin
    4  -rw-    13832032          <no date>  c1841-ipbase-mz.123-14.T7.bin
    2  -rw-       28282          <no date>  sigdef-category.xml
    1  -rw-      227537          <no date>  sigdef-default.xml

64016384 bytes total (16336765 bytes free)
R1# configure terminal
R1(config)# boot system flash:c1841-ipbase-mz.123-14.T7.bin
R1(config)# exit
R1# copy running-config startup-config
R1# reload
```

IOS Image Management

- To verify the primary boot image of a device, use the **show version** command

```
R1>show version
Cisco IOS Software, 1841 Software (C1841-IPBASE-M), Version 12.3(14)T7, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2006 by Cisco Systems, Inc.
Compiled Mon 15-May-06 14:54 by pt_team

ROM: System Bootstrap, Version 12.3(8r)T8, RELEASE SOFTWARE (fc1)

System returned to ROM by power-on
System image file is "flash:c1841-ipbase-mz.123-14.T7.bin"


This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
```

Password Recovery Procedures

- Passwords on devices are used to prevent unauthorized access.
- In case of a forgotten router or switch password, password recovery may be performed
- For encrypted passwords, such as the enable secret passwords, the passwords must be replaced after recovery.
- Depending on the device, the detailed procedure for password recovery varies; however, all the password recovery procedures follow the same general sequence:
  - **Step 1**. Connect to the device using a console cable
  - **Step 2.** Enter the ROMMON mode.
  - **Step 3**. Change the configuration register
  - **Step 4**. Copy the startup-config to the running-config.
  - **Step 5**. Change the password.
  - **Step 6**. Save the running-config as the new startup-config.
  - **Step 7**. Reload the device.

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Password Recovery Example

- **Step 1. Connect to the device using a console cable -** The password recovery procedure can be performed only through a console cable; hence physical access to the device is required

- **Step 2. Enter the ROMMon mode – ROM Monitor (ROMMon)** is a bootstrap program that initializes the device hardware (similar to a PC BIOS mode).
  - It is often used to perform various device recovery tasks which cannot be performed within the OS
  - The procedure to enter ROMMon varies among devices. For routers, it is accessed by pressing the CTRL-BREAK key sequence before the OS fully boots.
  - If successful, the prompt ‘rommon 1>’ appears

```
Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
############ <CTRL-BREAK>
monitor: command "boot" aborted due to user interrupt
rommon 1 >
```

Password Recovery Example

- **Step 3. Change the configuration register -** The configuration register is a setting that controls how a device boots depending on its value
  - **0x2142** = skip loading the startup config file during startup.
  - **0x2102** = load the startup config file during startup
  - Set it to ignore saved configurations using the command `confreg 0x2142` then boot up the router using the `reset` command

- **Step 4. Enter Privileged EXEC mode then copy the startup-config to the running-config**
  - This allows the original configuration to be loaded bypassing the prompt to enter the password
  - CAUTION: DO NOT copy running-config to startup-config before doing this step to avoid overwriting the saved device configuration with factory settings

```
monitor: command "boot" aborted due to user interrupt
rommon 1 > confreg 0x2142
rommon 2 > reset

System Bootstrap, Version 15.0(1r)M9, RELEASE SOFTWAR
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
...
 --- System Configuration Dialog ---

Would you like to enter the initial configuration
dialog? [yes/no]: n


Press RETURN to get started!


Router> enable
Router# copy startup-config running-config
Destination filename [running-config]?

613 bytes copied in 0.416 secs (1473 bytes/sec)
Router#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

Password Recovery Example

- **Step 5 Change the password. -** Since the device is currently in privileged EXEC mode, it is possible to enter Global configuration mode and replace all the necessary passwords.

- **Step 6. Save the running-config as the new startup-config** –
  - This allows the settings to be saved with the new passwords
  - Restore the configuration register setting using the command `config-register 0x2102` to allow the router to load the startup configuration the next time it boots

- **Step 7 (Optional). Reload the device**

```
613 bytes copied in 0.416 secs (1473 bytes/sec)
Router#
%SYS-5-CONFIG_I: Configured from console by console

R1# configure terminal
R1(config)# enable secret MyNewPass!



R1(config)# config-register 0x2102
R1(config)# end
R1# copy running-config startup-config
Destination filename [startup-config]? Building
configuration... [OK]









R1#reload
```

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What Did You Learn In This Module?

- Network management is an important aspect of administrating a network. It involves monitoring devices and managing device files
- Network discovery tools are used to gather the information about neighboring devices and are often used to map out a logical topology of a network when documentation is missing or lacking in detail.
  - Cisco Discovery Protocol (CDP) – Cisco proprietary
  - Link Layer Discovery Protocol (LLDP) – vendor neutral
- Event logging enables devices to notify an administrator about status changes and error events. Event timestamping is essential for an administrator to know when it occurred.
  - Network Time Protocol (NTP) is a time service that enables client devices to synchronize their system clocks with an NTP server
  - Syslog is a protocol that enables devices to log events to a buffer or a remote server for later analysis. Log entries include a facility code, severity code and an event description

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What Did You Learn In This Module?

- Simple Network Management Protocol (SNMP) is a protocol that allows devices to be centrally monitored and managed
  - A Network Management System (NMS) may use messages to request or change settings in a device hosting an SNMP agent
  - Device information and settings are stored in nodes within a hierarchical tree called the Management Information Base (MIB) are are accessed using their Object Identifiers (OIDs)
  - SNMP Traps are unsolicited messages that an SNMP agent can send to the NMS to report an event occurring on a device
  - SNMPv1 and SNMPv2c use ro and rw community strings to authenticate a NMS when it accesses the MIB
- Backing up device IOS images and configuration files is essential to allow error recovery. These may be backed up to:
  - Flash memory
  - USB drive
  - Remote server using Trivial File Transfer Protocol (TFTP)

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What Did You Learn In This Module?

- Password recovery procedures may be performed in case of forgotten passwords on routers or switches.
  - Requires physical access to the device using a console connection
  - Performed by entering ROMMon mode and setting the device to ignore the startup configuration when booting
  - Once booted into default settings, original configurations may be loaded back and the password replaced

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